

Conditional Generative Model

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Abstract

We investigate Conditional Generative Adversarial Networks (cGANs) for controllable human face generation using the CelebA dataset. By conditioning the generation process on selected facial attributes, the model learns to synthesize realistic face images that satisfy user-specified semantic constraints. We describe the rationale behind our attribute selection, the architecture of the proposed model, and the corresponding training objectives. Furthermore, we evaluate the effectiveness of the approach through both qualitative and quantitative experiments, assessing image realism, attribute consistency, and the model’s ability to generate diverse facial appearances. Our results demonstrate the effectiveness of conditional adversarial learning framework for controllable face synthesis.

1. Introduction

The task of generative face synthesis focuses on producing high-quality, realistic human face images. The development of deep generative models has significantly advanced this field, enabling not only unconditional face generation but also controllable synthesis conditioned on semantic attributes such as smiling expression, hair color, or the presence of eyeglasses. In this report, we investigate the use of Conditional Generative Adversarial Networks (cGANs) for attribute-guided face generation. Our approach leverages the CelebA dataset, a large-scale face dataset containing over 200,000 celebrity images annotated with multiple binary facial attributes. Using these annotations as conditioning signals, we train a model capable of generating faces that align with specified attribute combinations while maintaining visual realism.

2. Data Processing and Model

2.1. Selected Attributes and Data Processing

We condition our generative model on five binary facial attributes selected from the CelebA dataset: *Smiling*, *Eye-*

glasses, *Male*, *BlondHair*, and *Young*. This subset was chosen to provide a diverse range of statistical and visual characteristics, enabling a comprehensive evaluation of the model’s conditional generation capabilities. The selected attributes span both balanced and imbalanced distributions, allowing us to investigate the effect of class imbalance on conditioning performance. They also cover a broad range of spatial scales, from localized features such as smiles and eyeglasses to global facial characteristics such as age and gender, which influence overall facial structure and appearance.

In addition, the selected attributes exhibit varying degrees of correlation. In particular, combinations such as *Male* and *BlondHair* are relatively uncommon in the dataset, providing an opportunity to evaluate the model’s ability to generate rare attribute combinations without collapsing to dominant modes of the training distribution. At the same time, we avoid highly redundant attribute pairs whose co-occurrence patterns would overly constrain the conditioning space and limit the analysis of compositional generation.

Using five binary attributes results in 32 possible conditioning combinations. This conditioning space is sufficiently rich to study compositional generalization while remaining manageable for training and evaluation. The final attribute selection was guided by exploratory data analysis, including attribute distribution histograms and correlation heatmaps, which are provided in the `edaoutputs` directory.

To incorporate these conditioning signals, we implemented a custom dataset pipeline that follows the official CelebA training, validation, and testing partitions. For each sample, the dataloader outputs a tuple $(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} \in \mathbb{R}^{3 \times 64 \times 64}$ denotes the preprocessed image and $\mathbf{y} \in \{0, 1\}^5$ denotes the corresponding multi-hot conditioning vector.

The image preprocessing pipeline first center-crops each face image, then resizes it to 64×64 pixels. The resulting image is normalized to the range $[-1, 1]$. The attribute vector is constructed by converting CelebA’s original $(-1, 1)$ annotations into a standard binary representation, where 0

indicates the absence of an attribute and 1 indicates its presence.

2.2. Model Architecture

Conditional image generation requires a robust mechanism to synthesize samples that are both visually realistic and faithful to the conditioning signal. In this work, we adopt a Conditional Generative Adversarial Network (cGAN) structured under the classic Auxiliary Classifier GAN (AC-GAN) framework. Rather than employing Conditional Batch Normalization (CBN)—which modulates feature map statistics at every intermediate layer—we utilize a concatenation-based input conditioning strategy.

Our empirical observations indicated that CBN led to premature mode collapse. We hypothesize that modulating intermediate representations at every stage over-emphasizes the conditioning signal relative to the stochastic latent noise $\mathbf{z}$. This causes the generator to ignore the latent code in favor of a deterministic mapping of the labels. By switching to input-level concatenation, we establish a more balanced representation that preserves sample diversity while maintaining strong attribute control.

Generator: Latent Concatenation and Transposed Convolutions The generator G is designed to synthesize 64×64 RGB images from a combined latent and conditional representation. Let $\mathbf{z} \in \mathbb{R}^{128}$ be a latent noise vector drawn from a standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$, and let $\mathbf{y} \in \{0, 1\}^5$ represent the multi-label binary conditioning vector.

Conditioning is enforced at the input layer by directly concatenating the latent noise and the attribute label vector:

$$\mathbf{h}_0 = [\mathbf{z}; \mathbf{y}] \in \mathbb{R}^{133}$$

This concatenated vector is projected via a fully connected layer and reshaped into a spatial tensor of shape $(8 \times C_{\text{base}}) \times 4 \times 4$ (where $C_{\text{base}} = 64$).

Upsampling is performed progressively through a series of four transposed convolutional stages:

$$\text{Input } (4 \times 4) \rightarrow 8 \times 8 \rightarrow 16 \times 16 \rightarrow 32 \times 32 \rightarrow \text{Output } (64 \times 64)$$

Each upsampling block consists of a 2D fractional-strided convolution (kernel size 4, stride 2, padding 1), followed by 2D Batch Normalization and a Rectified Linear Unit (ReLU) activation function. The final layer maps the feature channels to the 3-channel RGB space using a convolutional layer capped with a hyperbolic tangent (tanh) activation function.

Discriminator: AC-GAN Backbone The discriminator D is structured as a standard convolutional neural network

that evaluates input images $\mathbf{x} \in \mathbb{R}^{3 \times 64 \times 64}$ without receiving the target labels as input, which is a key characteristic of the AC-GAN paradigm.

The image is processed through a downsampling backbone:

$$\text{Input } (64 \times 64) \rightarrow 32 \times 32 \rightarrow 16 \times 16 \rightarrow 8 \times 8 \rightarrow \text{Output } (4 \times 4)$$

consisting of standard 2D convolutions (kernel size 4, stride 2, padding 1) with LeakyReLU activations (slope $\alpha = 0.2$) and Batch Normalization (omitted in the first layer to stabilize initial gradient scaling).

After the final convolutional layer, the spatial feature map of shape $(8 \times C_{\text{base}}) \times 4 \times 4$ is flattened. The network then splits into two parallel, fully connected output heads:

1. **Adversarial Source Head (D_{source}):** Maps the flattened features to a scalar logit representing the probability that the input image is real rather than generated.
2. **Auxiliary Classification Head (D_{classes}):** Maps the same feature representation to $D = 5$ independent output logits, predicting the presence or absence of each conditioning attribute.

By decoupling the conditioning mechanism from intermediate layers and relying on dual classification objectives, this architecture ensures stable optimization and a clear mathematical separation between image realism and attribute faithfulness.

2.3. Training Objectives and Loss Functions

The optimization of our AC-GAN is formulated as a multi-objective minimax game. Rather than using Hinge Loss, both the adversarial validity and attribute conditioning are optimized using Binary Cross-Entropy with Logits. The total losses for the discriminator D and generator G are defined as:

$$\mathcal{L}_D = \mathcal{L}_D^{\text{adv}} + \lambda_{\text{cls}} \mathcal{L}_D^{\text{cls}}$$

where $\lambda_{\text{cls}} \geq 0$ is a hyperparameter scaling the auxiliary classification penalty (configured to 1.0 in our experiments).

Adversarial Source Loss To establish structural realism, the adversarial source head (D_{source}) is optimized to classify real images as 1 and fake (generated) images as 0. The adversarial loss components are defined as:

$$\begin{aligned} \mathcal{L}_D^{\text{adv}} &= -\mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log \sigma(D_{\text{source}}(\mathbf{x}))] \\ &\quad - \mathbb{E}_{\mathbf{z} \sim p_z, \mathbf{y} \sim p_y} [\log (1 - \sigma(D_{\text{source}}(G(\mathbf{z}, \mathbf{y}))))] \end{aligned}$$

$$\mathcal{L}_G^{\text{adv}} = -\mathbb{E}_{\mathbf{z} \sim p_z, \mathbf{y} \sim p_y} [\log \sigma(D_{\text{source}}(G(\mathbf{z}, \mathbf{y})))].$$

where $\sigma(\cdot)$ represents the sigmoid activation function. The generator is optimized to produce images that D_{source} classifies as real.

Auxiliary Attribute Classification Loss To enforce conditioning faithfulness, the auxiliary classification head (D_{classes}) outputs multi-label predictions which are evaluated against target attribute labels using independent binary cross-entropy.

* For the discriminator, $\mathcal{L}_D^{\text{cls}}$ is minimized solely on ****real**** images to train D_{classes} to be a highly accurate attribute classifier. * For the generator, $\mathcal{L}_G^{\text{cls}}$ is minimized on ****generated**** images, forcing G to synthesize the target traits specified in the input label vector y .

2.4. Evaluation and Results

The cGAN framework is evaluated through a comprehensive quantitative and qualitative pipeline designed to assess both semantic controllability and visual realism. Conditioning accuracy and F1-score are measured using an independent external convolutional classifier trained exclusively on real-data splits. The realism of generated samples is evaluated using the Fréchet Inception Distance (FID), which compares the feature distributions of real and generated images. Additionally, qualitative attribute sweeps are conducted by fixing latent vectors and individually toggling conditioning attributes to investigate feature disentanglement and identity preservation (Figure 1).

In 1, each pair of rows corresponds to turning a specific attribute off and on, respectively, for Smiling, Eyeglasses, Male, Blond Hair, and Young. The Smiling attribute demonstrates the strongest controllability, with the generator successfully modifying the mouth region while preserving the remaining facial identity. This behavior highlights the benefit of sufficient and balanced attribute representation in the training data. The Eyeglasses attribute exhibits weaker control due to its limited representation in the dataset, which leads to difficulties in learning this localized and less frequent feature. The Male attribute also shows strong semantic control, where non-male samples are consistently transformed into male appearances while already male identities remain unchanged, demonstrating the model’s ability to preserve existing attribute states. The Blond Hair attribute is also successfully manipulated, indicating that the generator has learned meaningful transformations in hair appearance. Finally, the Young attribute produces visible modifications in local facial characteristics, such as skin appearance and facial structure, although the changes are more subtle due to the diffuse nature of this attribute. Furthermore, 2 evaluates the compositional capability of the cGAN by examining combinations of multiple attributes rather than isolated feature changes. The rows correspond to individual and combined conditions,



Figure 1. Qualitative attribute sweep results. Each pair of rows shows the effect of disabling and enabling a single attribute while keeping the latent identity vector fixed. The evaluated attributes are Smiling, Eyeglasses, Male, Blond Hair, and Young.

including Smiling, Eyeglasses, Male, Blond Hair, Young, Smiling+Male, Blond Hair+Young, Blond Hair+Male, and Young Male with Eyeglasses. This experiment demonstrates the ability of the model to combine multiple semantic constraints and generate images with corresponding attribute interactions. As it can be seen in generation of Smiling+Male the model works better than Hair+Male and Young. Furthermore the model has tendency to generate women with blonde hair comparing to men with blonde hair. The model obtains a FID score of 23.85. This indicates that the generator successfully captures the complex facial appearance distribution of CelebA, including realistic facial structures, skin textures, eye regions, and hair details. Semantic controllability is particularly effective for attributes with strong visual signatures, such as Smiling and Male, achieving accuracies of 94.50% and 90.58% with F1-scores of 0.9413 and 0.9153, respectively. These attributes are well represented in the dataset and are therefore reliably controlled by the conditioning mechanism, producing smooth attribute transitions while maintaining identity consistency.

The model struggles with minority and subtle attributes due to dataset imbalance and bias. Eyeglasses represents the most significant failure case, achieving only 51.48% accuracy and an F1-score of 0.1133, indicating that the generator largely ignores this conditioning signal. Since eyeglasses are a rare attribute with highly localized geometry, the adversarial objective encourages the generator to prioritize global facial realism rather than accurately reproduc-



Figure 2. Compositional attribute generation results. The rows correspond to individual and combined conditioning vectors, including Smiling, Eyeglasses, Male, Blond Hair, Young, Smiling+Male, Blond Hair+Young, Blond Hair+Male, and Young Male with Eyeglasses. The results demonstrate the ability of the cGAN to generate images satisfying multiple semantic constraints simultaneously in case attributes have balance.

ing this minority feature. This also can be seen in figure1. As a result, the model achieves a strong FID score while sacrificing semantic accuracy for rare attributes. Similarly, Blond Hair suffers from insufficient representation in the dataset, causing the generator to favor more common hair distributions and resulting in a low F1-score of 0.2079. The Young attribute achieves relatively better performance with 67.34% accuracy and an F1-score of 0.7412; however, it remains challenging because youth-related characteristics are subtle, spatially distributed, and strongly correlated with identity-specific facial variations.